

WT-EXPERIMENT: 06

Topic: Water turbidity detector

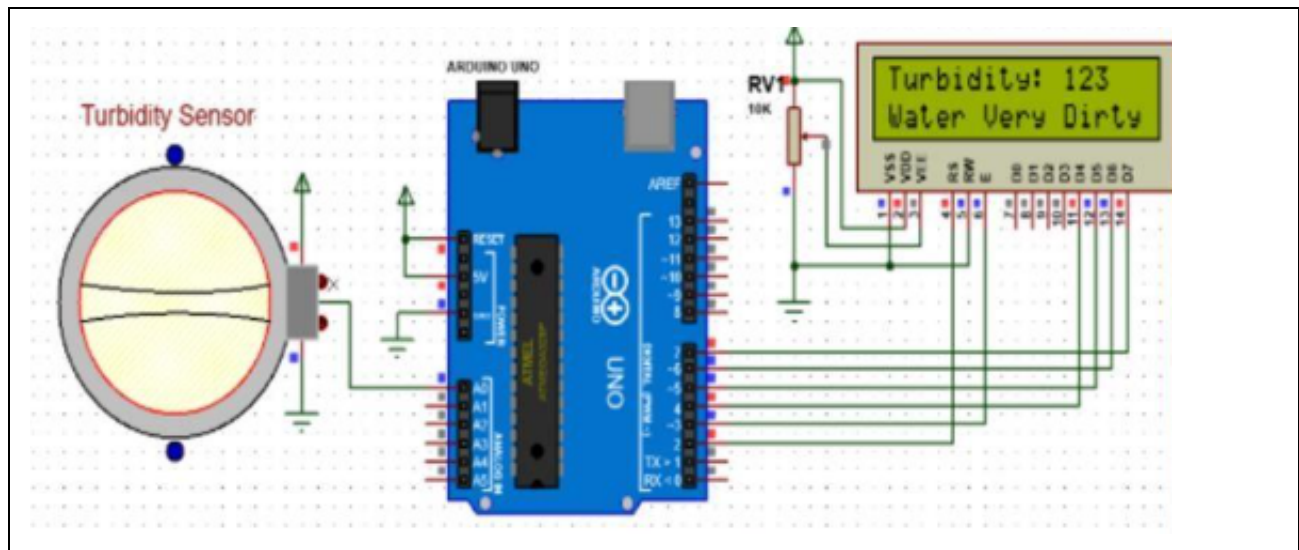
Aim:

To design and implement a water turbidity detector system using an Arduino UNO, a turbidity sensor, and an LCD display to measure and monitor the clarity of water in real-time.

Introduction:

Water turbidity detectors are essential tools for assessing water quality by measuring the cloudiness or haziness caused by suspended particles. Turbidity is a key indicator of contamination and is commonly monitored in drinking water systems, aquariums, rivers, and industrial applications.

This project utilizes an Arduino UNO and a turbidity sensor to detect and quantify the level of turbidity in water. The sensor measures the amount of light that passes through the water and is scattered by particles, providing an analog signal that corresponds to turbidity levels. The data is then processed by the Arduino and displayed on an LCD screen. This system allows for a quick, low-cost, and effective way to monitor water quality and detect possible contamination in real-time.



Components Used:

- Arduino UNO
- Turbidity Sensor (Analog)
- I2C LCD Display (16x2)
- Jumper Wires
- USB Cable (for power and code upload)

Connection of Components with Arduino UNO-Turbidity system:

Component	Pin on Component	Connected to Arduino UNO
Turbidity Sensor	VCC	5V
	GND	GND
	Signal (Analog Output)	A0 (Analog Pin)
I2C LCD Display	VCC	5V
	GND	GND
	SDA	A4
	SCL	A5
Power Supply	USB Cable	USB Port (for powering and uploading)

Code for Connection:

```
#include
<LiquidCrystal_I2C.h>

LiquidCrystal_I2C lcd(0x27,
2, 16);
int sensorPin = A0;

void setup() {
  Serial.begin(9600);
  lcd.init();
  lcd.backlight();
}

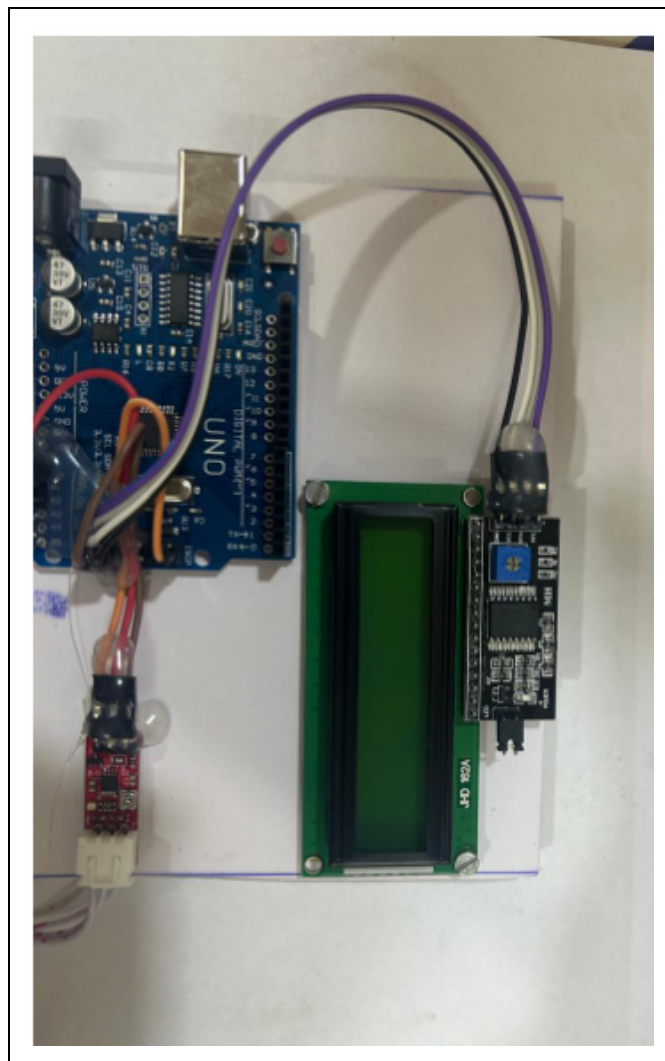
void loop() {
  int sensorValue =
analogRead(sensorPin);
  Serial.print("Sensor Value: ");
  Serial.println(sensorValue);
```

```
int turbidity =  
map(sensorValue, 0, 520, 100,  
0);  
  
lcd.setCursor(0, 0);  
lcd.print("Turbidity:  ");  
lcd.setCursor(10, 0);  
lcd.print(turbidity);  
  
lcd.setCursor(0, 1);  
if (turbidity < 20) {  
    lcd.print("It's CLEAR  ");  
    Serial.println("It's  
CLEAR");  
} else if (turbidity >= 20 &&  
turbidity < 50) {  
    lcd.print("It's CLOUDY  
");  
    Serial.println("It's  
CLOUDY");  
} else {  
    lcd.print("It's DIRTY  ");  
    Serial.println("It's DIRTY");  
}  
  
delay(1000);  
}
```

Libraries Used:

- LiquidCrystal_I2C.h: Used to control the I2C LCD display for showing turbidity levels and water quality status.
- No additional libraries are needed as the turbidity sensor provides analog output that can be directly read using `analogRead()`.

Implementation:



- Connect the turbidity sensor and LCD display to the Arduino as per the wiring configuration.
- Upload the turbidity monitoring code to the Arduino UNO via USB.
- Open the Serial Monitor (set baud rate to 9600).
- Submerge the sensor in water to measure turbidity.
- The LCD displays the turbidity value and corresponding water quality status (e.g., CLEAR, CLOUDY).
- The turbidity readings are also printed to the Serial Monitor for real-time debugging or observation.

Conclusion:

This project effectively demonstrates a simple and efficient turbidity detection system using Arduino. By utilizing an analog turbidity sensor and displaying real-time readings on an I2C LCD display, the system offers a convenient method to assess water clarity. It categorizes water quality based on turbidity levels, indicating conditions such as CLEAR or CLOUDY. This system can be practically applied in various environments such as water treatment plants, environmental monitoring, and household water quality analysis, promoting awareness and safety in water usage.